

1 Definitions

For a function $f : [n] \rightarrow \mathbb{R}$ we say a pair $(i, i + 1)$ for i in $[n - 1]$ is

1. a *good pair* if there exists an optimal LIS that does not change the values in this pair
2. a *bad pair* if no optimal LIS includes this pair untouched
3. a *perfect pair* is a pair that must remain untouched in all optimal LIS

Note that all $(j, j + 1)$ such that $f(j) > f(j + 1)$ are "bad pairs". Also note that all perfect pairs are also good pairs

2 $1 + \delta$ assuming $\varepsilon_f < 0.5$

Theorem 2.1. *For a fixed delta, if the true ε_f is less than 0.5, Using distance-separation(f, ε, δ), the decision problem of if f is within $\varepsilon \pm \delta$ is solvable in polylog time.*

3 good pair analysis

3.1 fraction of good pairs

Theorem 3.1. *At least half of all increasing pairs are good pairs*

3.2 padding good pairs

Theorem 3.2. *given a good pair $(i, i + 1)$, constructing a new function $f_p : [n + 1] \rightarrow \mathbb{R}$ were*

$$f_p(x) = \begin{cases} f(x), & x \leq i \\ \frac{f(i) + f(i+1)}{2}, & x = i + 1 \\ f(x + 1), & x \geq i + 2 \end{cases} \quad (1)$$

forces $(i, i + 1), (i + 1, i + 2)$ to both be perfect pairs in the new function f_p